KIERAN PEREIRA

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EDUCATION

University of California, Berkeley, USA **August 2024 – May 2025**

MEng Mechanical Engineering, Concentration in Robotics

Capstone Project: *CaptAIn - autonomous sailboat fleet for ocean data capture*

- Owned the control system and integrated the hardware, firmware and software for a fleet designed to scale to 1,000 units. Shipped the controls in ROS2/C++ after evaluating PID, LQR and MPC.
- Built state-estimation software (a 1-D Kalman filter) that cut sensor noise by 85%, and a replay-based hardware-in-the-loop simulator that cut iteration time from days to hours.
- Field-tested 18 of the 50+ boats built, completing a 2.3 km fully autonomous upwind and downwind run on SF Bay.

University College London (UCL), London, UK **September 2020 – 2024**

BEng Mechanical Engineering with Control for Intelligent Systems – First Class Honors (4.0 GPA)

- Capstone Project (4.0 GPA). Real-time traffic-sign detection and narration for new drivers (PyTorch/YOLOv5, 100k images, 0.81 F1, ~15 FPS).

PROFESSIONAL EXPERIENCE

Undaunted Robotics, Head of Robotics – Founding Engineer, Atlanta, USA **July 2025 – Present**

- Led the hardware stack as the fleet grew from 10 to 150 robots, now the largest robotic security fleet in the US, including charging infrastructure, in-house payloads, a custom rover and vendor-independent fleet control.
- Proposed a security rover and led the 3-person team to build it. Owned the mechanical design, power system and motor-control firmware. The first unit field-deployed in 2 months, and the fleet has grown to 20 units at 96% uptime, responding to 130 incidents. Built DFM partnerships with US and international manufacturers, cutting unit cost by 60%.
- Architected a custom motor-driven steering column for the rover, with a reduction stage feeding a CNC-machined column, to replace dual linear-actuator steering. Leading CAD and prototyping.
- Designed the production charging hub. There are 140 hubs in the field, built at 10 a week by local manufacturers. Heat-related outages stayed below 3% over 12 months, and the hub unblocked ~$620K in stalled deals.
- Diagnosed a single-vendor dependency after a cloud outage took the entire fleet offline. Architected control independent of the vendor's cloud and shipped it as a 5-minute-per-robot field upgrade in 1 week.
- Integrated cameras, audio and lighting into an in-house payload that replaced an off-the-shelf camera. Built the Raspberry Pi teleop stack that streams live video and two-way audio to the customer dashboard.

Lockheed Martin, Systems Engineer, UK **June 2023 – June 2024**

- Owned system requirements for a safety-critical radar system, translating customer and manufacturer needs into 600+ traceable design requirements for cross-functional engineering teams.
- Built and ran software-in-the-loop tests to verify radar system performance, simulating communications across 200+ entities to validate system behavior ahead of integration.
- Led an object-detection R&D initiative to strengthen QA, presenting the engineering and financial case to senior executives and securing $104K in pilot funding.

LEADERSHIP EXPERIENCE

Co-Founder, Obsidian- Full Motion Tracking Wearables, Berkeley, USA **August 2024 – May 2025**

- Co-founded a motion-tracking wearable startup and led a 5-person team from concept to working prototype. Validated demand through 156 customer interviews (NSF I-Corps) and secured 2 letters of intent.
- Built the embedded system on an ESP32 running FreeRTOS (3 concurrent tasks) integrating two IMUs, a barometer and wireless connectivity.
- Wrote Kalman-filter sensor fusion that reached 95% precision against measured movement.

Head of Payload Subsystem, MACH22 National Rocketry Competition, London– 2nd / 32 **September 2022 – May 2023**

- Led a 5-engineer payload team to design, build, and fly a CanSat to 1.5 km with successful parachute recovery.
- Co-founded UCL Rocketry; secured sponsorship and helped grow it into UCL's strongest competition engineering team.
- Owned CAD and deployment mechanisms for parachute and 360° camera systems into a 3D-printed CanSat shell.

RELEVANT SKILLS

Mechanical: *Fusion 360, DFM, rapid prototyping, 3D printing*
Software: *Python, C++, MATLAB/Simulink/Simscape, Git, Linux, PyTorch*
Test & Validation: *HIL/SIL testing, field testing, data logging and analysis, root-cause debugging, requirements traceability*
Electrical & Embedded: *BLDC motor control, wiring harnesses, power systems, CAN/I2C/UART/SPI, ESP32/FreeRTOS, Raspberry Pi*
Sensing: *IMUs, cameras, GPS, strain gauges, barometers, Kalman filtering, sensor fusion*
Robotics & Controls: *ROS2, PID/LQR/MPC, state estimation, teleoperation (MQTT, LTE)*